

Boolean Algebra

$$(B, +, \cdot, ', I, 0)$$

- 1) Commutative property: $x+y = y+x$, $x \cdot y = y \cdot x$,
 $\forall x, y \in B$.
- 2) Distributive property: $x \cdot (y+z) = x \cdot y + x \cdot z$
 $x + yz = (x+y)(x+z)$
 $\forall x, y, z \in B$
- 3) Identity Law: $a+0 = 0+a = a$
 $a \cdot 1 = a = 1 \cdot a$
- 4) Complement Law: $a+a' = 1$, $a \cdot a' = 0$.

In Boolean Algebra,

- i) Additive Identity if exists is unique.
- ii) Multiplicative " " " "
- iii) Complement of each element if exists is unique.

- 5) Idempotent Law: (i) $a+a = a$ (ii) $a \cdot a = a$
 $\forall a \in B$

Proof: i) $(a+1) = (a+1) \cdot 1$
 $= (a+1) \cdot (a+a')$
 $= a + a \cdot a' = a + 0 = a$

ii) $a \cdot a = a + 0$
 $= a + a \cdot a'$
 $= a \cdot (a+a') = a \cdot 1 = a$

For $a, b \in B$

ci) $a+1=1$ cii) $a \cdot 0 = 0$ ciii) $a+ab=a$
 civ) $a \cdot (a+b) = a$

i) $a+1 = (a+1) \cdot 1 = (a+1)(a+a') = a + 1 \cdot a' = a + a' = 1$

ii) $a \cdot 0 = (a \cdot 0) + 0 = (a \cdot 0) + (a \cdot a')$
 $= a(0+a') = a(a') = 0$

iii) $a + a \cdot b = a \cdot 1 + ab = a(1+b) = a \cdot 1 = a$

iv) $a(a+b) = (a+0)(a+b) = a + (0 \cdot b) = a + 0 = a$

In boolean algebra B ,

a) $0' = 1$ (2) $1' = 0$ (3) $(a+b)' = a'b'$ (4) $(ab)' = a'+b'$

$$\text{of (i)} \quad 0' = 0' + 0 = 1 \quad \text{C. } a + a' = 1$$

$$\text{ii)} \quad 1' = 1' \cdot 1 = 0 \quad \text{C. } a \cdot a' = 0$$

$$\text{iii)} \quad \text{To prove that: } (a+b) + a'b' = 1 \approx (a+b)' = a'b'$$

$$\underbrace{(a+b)}_{\equiv 1} + a'b' \Rightarrow 1 + a'b'$$

$$(1+a') \cdot (1+b')$$

$$= (a+b+a')(a+b+b') = (1+b) \cdot (a+1) = 1 \cdot 1 = 1$$

$$(a+b) \cdot \underbrace{(a'b')}_{\equiv 0} \Rightarrow a \cdot 0 + b \cdot 0$$

$$= a(a'b') + b(a'b') = (a \cdot a')b' + (b \cdot b')a' = 0 + 0 = 0$$

For any $a, b \in B$

$$(1) \quad a+b=b \Leftrightarrow a'+b=1$$

$$(2) \quad a'+b=1 \Rightarrow ab'=0$$

$$(1) \quad a+b=b \Rightarrow a'+a+b = a'+b$$

$$1+b = a'+b$$

$$1 = a'+b$$

$$\begin{aligned} &= \cancel{(a+b)(a'+b) = (a+b)} \\ &\quad \cancel{a \cdot a' + b = a+b} \end{aligned}$$

$$(2) \quad (a'+b)' = 1' \Rightarrow (a')' \cdot b' = 0 \Rightarrow a \cdot b' = 0$$

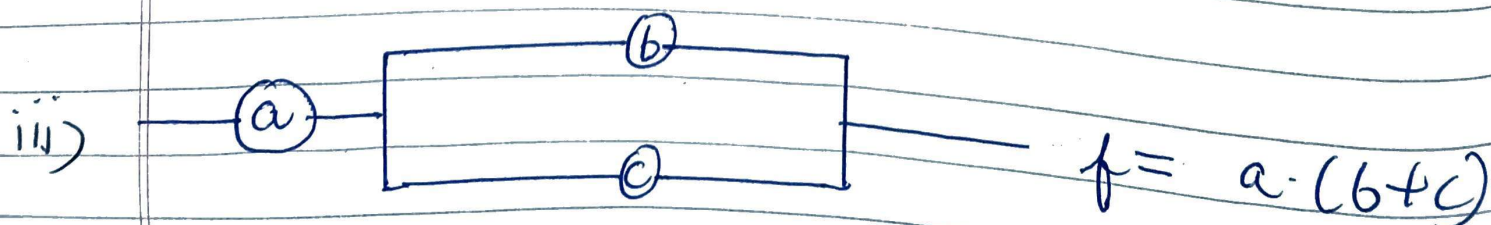
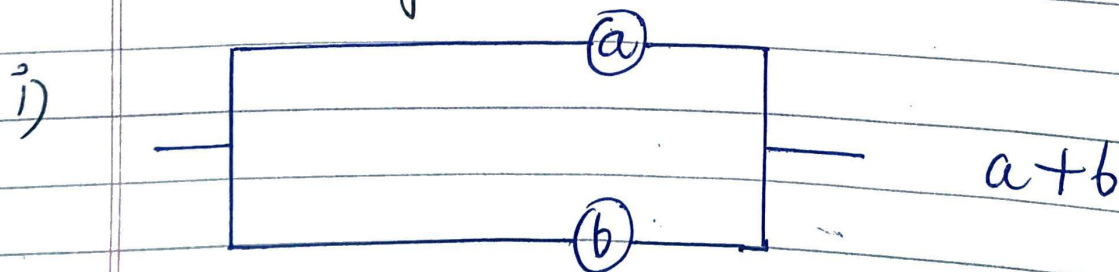
Q. Using Boolean Algebra. Prove that
(i) $a \cdot b + b \cdot c + b \cdot c' = b$

$$\begin{aligned} &\Rightarrow a \cdot b + b \cdot (c + c') \\ &= a \cdot b + b \cdot 1 \\ &= a(b + 1) \\ &= b \end{aligned}$$

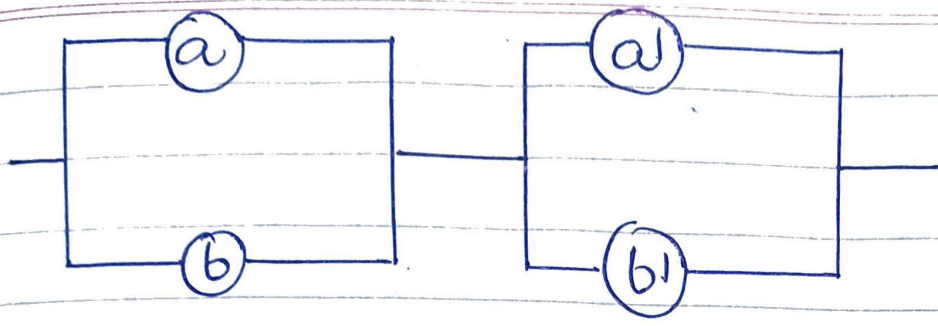
$$\begin{aligned} \text{(ii)} \quad &abc + a' + b' + c' = 1 \\ &= abc + (abc)' = 1 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad &(a+b)(a'+c) = ac + a'b + bc \\ &= (a+b)a' + (a+b)c \quad [\text{Distributive property}] \\ &= a \cdot a' + a'b + ac + bc \\ &= 0 + a'b + ac + bc \quad [aa' = 0] \\ &= a'b + ac + bc \quad [0 + a = a] \end{aligned}$$

Switching Circuits



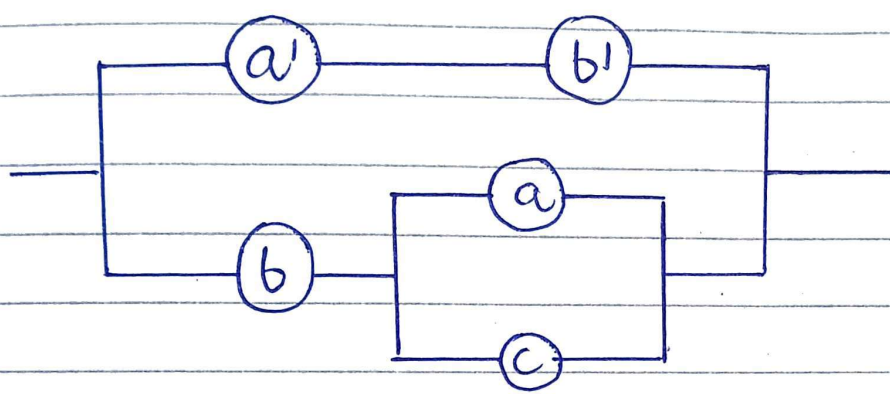
Q. 1)



$$f = (a + b) \cdot (a' + b')$$

Q. 2)

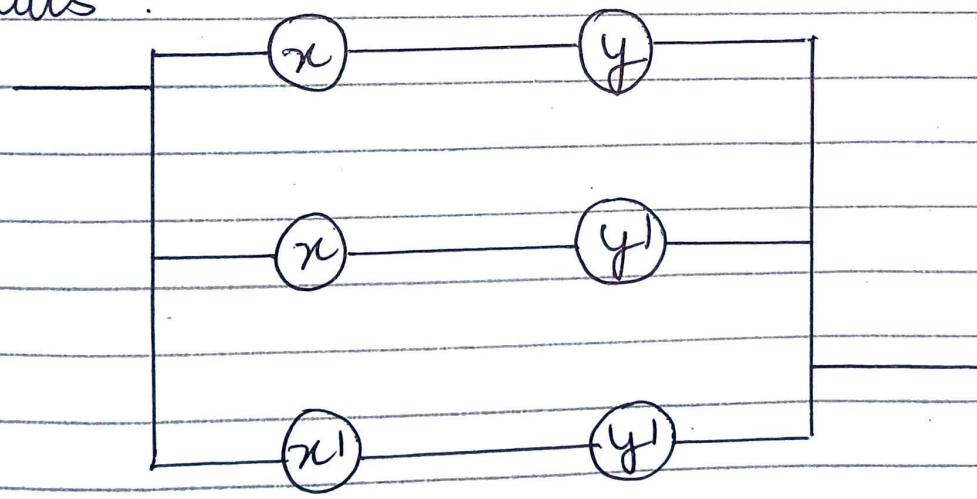
$$f = a' \cdot b' + b \cdot (a + c)$$



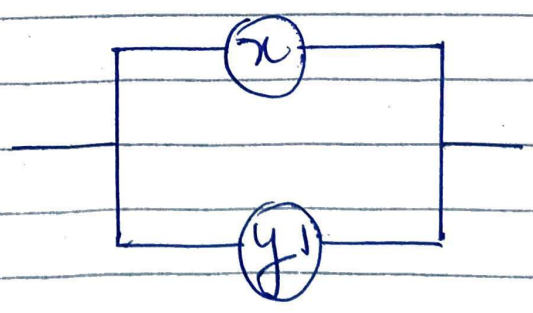
Q. 3)

Draw simpler circuits by simplifying boolean circuits :

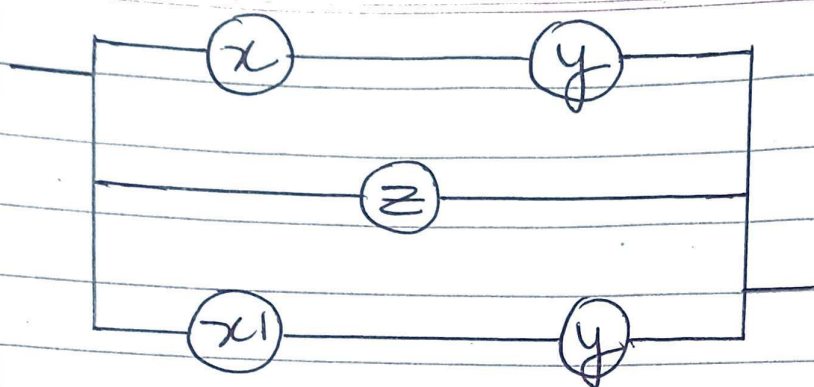
Q. 4)



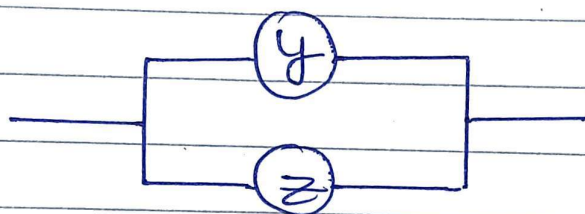
$$\begin{aligned} &\Rightarrow xy + xy' + x'y' \\ &= xy + (x + x')y' \\ &= xy + y' \\ &= (x + y')(y + y') \\ &= (x + y') \cdot 1 \\ &= x + y' \end{aligned}$$



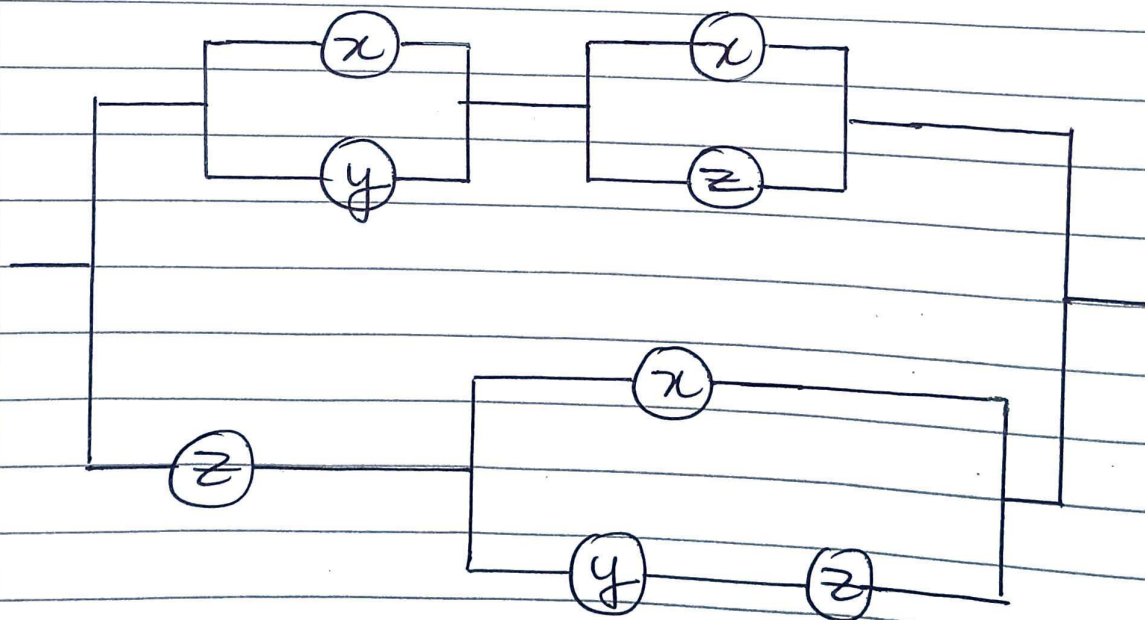
Cii)



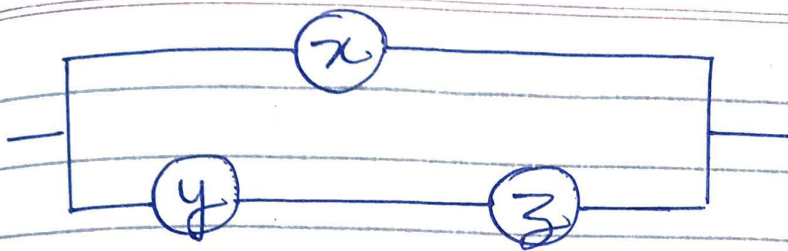
$$\begin{aligned}
 &= x \cdot y + z + x \cdot y \\
 &= (x + x) \cdot y + z \\
 &= 1 \cdot y + z = y + z
 \end{aligned}$$



Ciii)

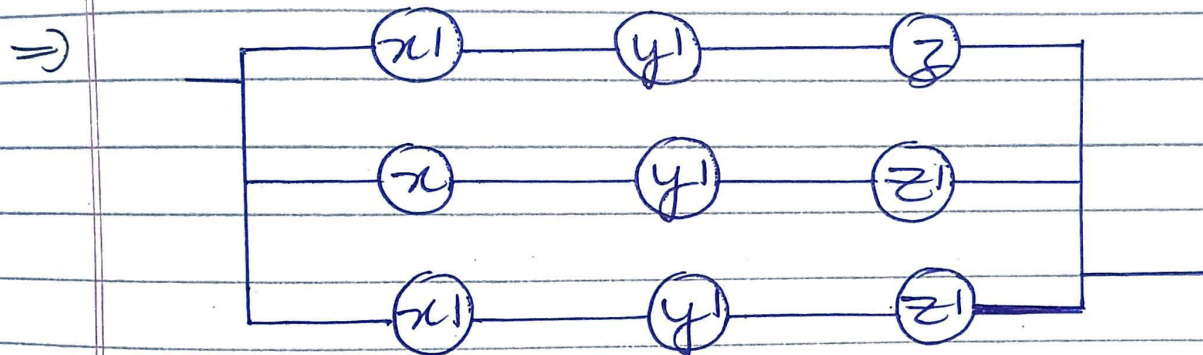


$$\begin{aligned}
 f &= (x + y) \cdot (x + z) + z \cdot (x + y \cdot z) \\
 &= (x + y \cdot z) + z \cdot (x + y \cdot z) \\
 &= (x + y \cdot z) (1 + z) \\
 &= x + y \cdot z
 \end{aligned}$$

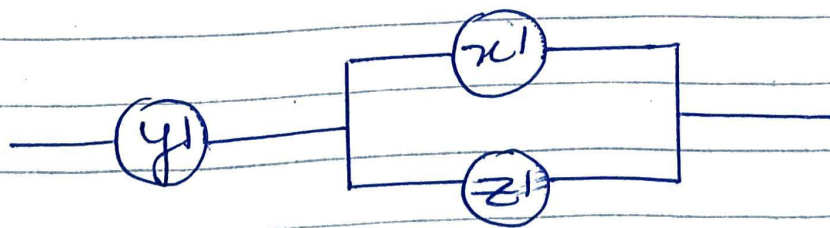


Q. Draw switching circuit of boolean function f , simplify it and draw the switching circuit of the simplified version.

$$f = x'y'z + xy'z' + x'y'z'$$



$$\begin{aligned}
 f &= x'y'z + xy'z' + x'y'z' \\
 &= (x' + x)y'z' \\
 &= x'y'z' + (x + x')y'z' \\
 &= x'y'z' + 1 \cdot y'z' \\
 &= (x'z + z')y' \\
 &= [(x' + z) \cdot (z + z')] \cdot y' \\
 &= (x' + z) \cdot y'
 \end{aligned}$$



Karnaugh Map (K-Map)

2 variables (x, y)

$$2^2 = 4$$

$x \backslash y$	y'	y
x'	0	1
x	2	3

3 variables (x, y, z)

$$2^3 = 8$$

$x \backslash yz$	$y'z'$	$y'z$	yz	yz'
x'	0	1	3	2
x	4	5	7	6

Q. Simplify the following using Karnaugh map.

(a)

$$F = A'B'C' + A'B'C$$

$A \backslash BC$	$B'C'$	$B'C$	BC	BC'
A'	1			
A	1			

$$\Rightarrow B'C'$$

b) $f = A'B'C' + AB'C' + AB'C + A'BC' + ABC'$

$A \backslash BC$	$B'C'$	$B'C$	BC	BC'
A'	1			1
A	1	1		1

$$\Rightarrow C' + AB'$$

c) $f = A'B'C' + A'B'C + A'BC + AB'C + ABC'$

$A \backslash BC$	$B'C'$	$B'C$	BC	BC'
A'	1	1	1	
A		1		1

$$\Rightarrow A'B' + A'C + B'C + ABC'$$

d) $f = A'B'C' + A'BC' + AB'C + ABC$

$A \backslash BC$	$B'C'$	$B'C$	BC	BC'
A'	1			1
A		1	1	

$$\Rightarrow A'C' + AC = A \odot C$$